Seth Ricks ECEN 380 3/17/2026

PS10: 6.27(a), 6.29(b), 6.36(h), 6.37(d), 6.38(c), MATLAB 6.66(a,b,c), 6.67(a,b), {6.31 → 6.35a}

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6.27a Determine the bilateral Laplace transform and ROC for the following signal:

(a) $x(t) = e^{-t}u(t+2)$

$$x(t) = e^{-t}u(t+2)$$

$$x(t) = \frac{e^{-2}}{e^{-2}} e^{-t}u(t+2) = \frac{1}{e^{-2}} e^{-t+2} u(t+2) = e^{2} e^{-(t+2)} u(t+2)$$

D.1:

$x(t) = e^{-at}u(t) \overset{\mathcal{L}}{\longleftrightarrow} X(s) = \frac{1}{s+a}$; ROC: $Re\{s\} > -a$

$b(t) = e^{-t}u(t)$, where $x(t) = e^{2}b(t+2)$

$B(s) = \frac{1}{s+a} = \frac{1}{s+1}$, ROC: $Re\{s\} > -1$

D.2:

$x(t-\tau) \longleftrightarrow e^{-s\tau}X(s)$; ROC $= R_x$

$b(t+2) = b(t--2) = e^{-s(-2)}(s)$; ROC $= R_x$

$= e^{2s}B(s)$; ROC $= R_x$

$= e^{2s}\left(\frac{1}{s+1}\right)$; ROC $= R_x$

$$X(s) = \mathcal{L}\{e^{2}b(t+2)\} = e^{2}\mathcal{L}\{b(t+2)\} = e^{2}\cdot e^{2s}\cdot\left(\frac{1}{s+1}\right);$$

ROC $= R_x$

$$\boxed{X(s) = \frac{e^{2s+2}}{s+1}; \text{ ROC: } Re\{s\} > -1}$$

6.29(b) Use the basic Laplace transforms and the Laplace transform properties given in Table D.1 and D.2 to determine the unilateral Laplace transform of the following signals:

$$x(t) = t\,u(t) * \cos(2\pi t)\,u(t)$$

D.1:

$$\mathcal{L}\{t\,u(t)\} = \frac{1}{s^2}, \quad \text{ROC: } Re\{s\} > 0$$

$$\mathcal{L}\{[\cos(\omega_1 t)]u(t)\} = \frac{s}{s^2+\omega_1^2}, \quad \text{ROC: } Re\{s\} > 0$$

D2:

$$\mathcal{L}\{x(t) * y(t)\} = X(s)Y(s) \quad \text{if } x(t)=y(t)=0 \text{ for } t<0, \quad \text{ROC: at least } R_x \cap R_y$$

$$b(t) = t\,u(t) \rightarrow B(s) = \frac{1}{s^2}, \quad \text{ROC: } Re\{s\} > 0$$

$$c(t) = \cos(2\pi t)\,u(t) \rightarrow C(s) = \frac{s}{s^2+(2\pi)^2}, \quad \text{ROC: } Re\{s\} > 0$$

$$x(t) = b(t) * c(t) \rightarrow X(s) = B(s)C(s) = \left(\frac{1}{s^2}\right)\left(\frac{s}{s^2+(2\pi)^2}\right),$$

$$\boxed{X(s) = \frac{1}{s(s^2+4\pi^2)}, \quad \text{ROC: } Re\{s\} > 0}$$

0.3(b) Determine the final value $x(\infty)$, given the following Laplace transform $X(s)$:

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$$X(s) = \frac{s+2}{s^3+2s^2+s}$$

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$$X(s) = \frac{s+2}{s^3+2s^2+s} = \frac{s+2}{s(s^2+2s+1)} = \frac{s+2}{s(s+1)^2}$$

$$= \frac{A}{s} + \frac{B}{(s+1)} + \frac{C}{(s+1)^2}$$

$$s+2 = A(s+1)^2 + B(s)(s+1) + C(s)$$

$$= A(s^2+2s+1) + Bs^2 + Bs + Cs$$

$$= As^2 + 2As + A + Bs^2 + Bs + Cs$$

$$\begin{aligned} A+B&=0 \\ 2A+B+C&=1 \\ A&=2 \end{aligned} \longrightarrow \begin{aligned} 2+B&=0 \\ 4+B+C&=1 \\ A&=2 \end{aligned} \longrightarrow \begin{aligned} B&=-2 \\ 2+C&=1 \\ A&=2 \end{aligned} \longrightarrow \begin{aligned} B&=-2 \\ C&=-1 \\ A&=2 \end{aligned}$$

$$X(s) = \frac{2}{s} + \frac{-2}{s+1} + \frac{-1}{(s+1)^2}$$

D.1: $\mathcal{L}\{u(t)\} = 1/s$, $ROC = Re\{s\} > 0$

$\mathcal{L}\{e^{-at}u(t)\} = \frac{1}{s+a}$, $ROC = Re\{s\} > -a$

$\mathcal{L}\{te^{-at}u(t)\} = \frac{1}{(s+a)^2}$, $ROC = Re\{s\} > -a$

$\mathcal{L}\{2/s\} = 2u(t)$

$\mathcal{L}\{\frac{-2}{s+1}\} = -2e^{-t}u(t)$

$\mathcal{L}\{\frac{-1}{(s+1)^2}\} = -te^{-t}u(t)$

$x(t) = [2 - 2e^{-t} - te^{-t}]u(t)$, $ROC = Re\{s\} > 0$

$x(\infty) = [2 - 2e^{-\infty} - \infty e^{-\infty}]$; $\lim_{t\to\infty} te^{-t} = \lim_{t\to\infty} \frac{t}{e^t} = \lim_{t\to\infty} \frac{1}{e^t} = 0$

$$\boxed{x(\infty) = 2}$$

6.37(d) Use partial fractions to find the time signal corresponding to unilateral Laplace Transform:

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(d) $X(s) = \dfrac{5s+4}{s^3+3s^2+2s}$

$$X(s) = \frac{5s+4}{s(s^2+3s+2)} = \frac{5s+4}{s(s+2)(s+1)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+1}$$

$$5s+4 = A(s+2)(s+1) + B(s)(s+1) + C(s)(s+2)$$
$$= A(s^2+3s+2) + Bs^2 + Bs + Cs^2 + 2Cs$$

$$= As^2 + 3As + 2A$$
$$+ Bs^2 + Bs$$
$$+ Cs^2 + 2sC$$

$$A+B+C=0$$
$$3A+B+2C=5$$
$$2A+0B+0C=4$$

$$\xrightarrow{RREF} \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$X(s) = \frac{2}{s} + \frac{-3}{s+2} + \frac{1}{s+1}$$

D.1: $\mathcal{L}\{u(t)\} = 1/s$, $ROC = Re\{s\} > 0$

$\mathcal{L}\{e^{-at}u(t)\} = \frac{1}{s+a}$, $ROC = Re\{s\} > -a$

$\mathcal{L}\{te^{-at}u(t)\} = \frac{1}{(s+a)^2}$, $ROC = Re\{s\} > -a$

$x(t) = 2u(t) + -3e^{-2t}u(t) + e^{-t}u(t)$, $ROC = Re\{s\} > 0$

$$\boxed{x(t) = [2 - 3e^{-3t} + e^{-t}]\,u(t),\ ROC = Re\{s\} > 0}$$

6.38(c) Determine the forced and natural responses for the LTI system described by the following differential equations w/ specified input & initial conditions:

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(c) $\frac{d^2}{dt^2} y(t) + y(t) = 8x(t),\ y(0^-) = 0,$

$\frac{d}{dt} y(t)\Big|_{t=0^-} = 2,\ x(t) = e^{-t}u(t)$

D.2: $\mathcal{L}\left\{\frac{d}{dt}x(t)\right\} = sX(s) - x(0^-)$, ROC = at least Rx

D.1: $\mathcal{L}\{e^{-at}u(t)\} = \frac{1}{s+a}$, ROC = $Re\{s\} > -a$

$\mathcal{L}\left\{\frac{d}{dt}y(t)\right\} = \underbrace{sY(s) - y(0^-)}$

$\mathcal{L}\left\{\frac{d^2}{dt^2}y(t)\right\} = s(\ ) - \frac{d}{dt}y(t)\Big|_{t=0^-}$

$= s^2Y(s) - sy(0^-) - \frac{d}{dt}y(t)\Big|_{t=0^-}$

$s^2Y(s) - sy(0^-) - \frac{d}{dt}y(t)\Big|_{t=0^-} = 8X(s)$

$X(s) = \mathcal{L}\{e^{-t}u(t)\} = \frac{1}{s+1}$, ROC = $Re\{s\} > -a$

$s^2Y(s) - sy(0^-) - \frac{d}{dt}y(t)\Big|_{t=0^-} = \frac{8}{s+1}$

$$Y(s) = \underbrace{\frac{8}{(s+1)s^2}}_{f} + \underbrace{\frac{\frac{d}{dt}y(t)\Big|_{t=0^-} + sy(0^-)}{s^2}}_{n}$$

$Y^{(f)}(s) = \frac{8}{s^2(s+1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+1}$ $\quad Y^{(n)}(s) = \frac{2 + 0s}{s^2} = \frac{2}{s^2}$

$8 = A(s^2)(s+1) + B(s)(s+1) + C(s^3)$

$8 = As^3 + As^2 + Bs^2 + Bs + Cs^3$

$\boxed{y^{(n)}(t) = 2u(t),\ ROC = Re\{s\} > 0}$

$Y^{(f)}(s) = \frac{-8}{s} + \frac{8}{s^2} + \frac{8}{s+1}$

$\boxed{y^{(f)}(t) = -8u(t) + 8tu(t) + e^{-t}u(t),\ ROC = Re\{s\} > 0}$